

PREDICTIVE MAINTENANCE OF VEHICLES USING MACHINE LEARNING

LITERATURE REVIEW:

Objective of the Literature Survey

To review recent machine learning approaches for predictive vehicle maintenance, identify commonly used algorithms and methodologies, analyze their strengths and limitations, and establish the motivation for the proposed logistics fleet vehicle maintenance framework.

Paper 1: A Comprehensive Study on Predicting the Need for Vehicle Maintenance Using Machine Learning

Objective

Develop a machine learning framework to predict maintenance requirements using ML.

Literature Survey

The study applies SMOTE and compares Decision Tree, Random Forest, Gradient Boosting, Naive Bayes and KNN for maintenance prediction. It shows ML can reduce downtime and maintenance cost.

Paper 2: Machine Learning Applications in Predictive Maintenance for Vehicles: Case Studies

Objective

Review ML applications in vehicle predictive maintenance.

Literature Survey

The paper surveys ML algorithms, IoT sensors and case studies across vehicle subsystems, providing an overview of predictive maintenance practices.

Paper 3: Predictive Maintenance Enabled by Machine Learning: Use Cases and Challenges in the Automotive Industry

Objective

Survey ML-enabled predictive maintenance in automotive systems.

Literature Survey

The authors categorize automotive PdM applications, ML methods and future challenges including data availability and explainability.

Paper 4: Predictive Maintenance in Industry 4.0: A Survey of Planning Models and Machine Learning Techniques

Objective

Review planning models and ML techniques for Industry 4.0 PdM.

Literature Survey

The paper discusses data cleaning, normalization, feature selection, decision models and prediction models for industrial predictive maintenance.

Paper 5: Predictive Maintenance of Armoured Vehicles using Machine Learning Approaches

Objective

Develop an ensemble ML model for armoured vehicles.

Literature Survey

The paper evaluates ensemble models with cross-validation to predict maintenance needs of defence vehicles.

Conclusion

The reviewed papers confirm the effectiveness of machine learning for predictive maintenance while highlighting challenges such as limited public datasets, insufficient feature engineering, lack of feature leakage analysis, and limited logistics fleet validation. These findings motivate the proposed research.